

# EduVision: Higher Education Performance Dashboard Suite

## Executive Summary

The **EduVision Higher Education Performance Dashboard** is a 4-tab interactive visual analytics suite built in Tableau. It provides comprehensive insights into global university rankings, research outputs, student body demographics, and institutional metrics across 106 countries.

Designed with an executive slate, unified KPI summary cards, and dynamic navigation, the dashboard suite empowers academic leaders and stakeholders to evaluate performance without data inflation.

## Repository & Module Structure

**springboardmentorpl/Higher-Education-Performance-Dashboard-Group-1/ (Branch: Quainaat\_Rizvi)**

```
|
|
| — Module_1_Data_Collection/
|   | — data_collection.py.py
|   | — university_raw_data.csv
|
| — Module_2_Data_Cleaning/
|   | — education_cleaning.ipynb
|   | — university_cleaned.csv
|
| — Module_3_Dataset_Finalization/
|   | — generate_education_kpis.py
|   | — university_final_dataset.xlsx
|
| — Module_4_Wireframing/
|   | — dashboard_storyboard.pdf
|
| — Module_5_Tableau_Prototypes/
|   | — eduvision_prototype.twbx
```

```
|
|
|— Module_6_Dashboard_Development/
|   |— calculated_fields_summary.md
|
|
|— Module_7_Interactivity_Testing/
|   |— dashboard_testing_matrix.pdf
|
|
|— Module_8_Final_Submission/
|   |— README.md
|   |— eduvision_dashboard_v1.twbx
```

## Dashboard Architecture

The suite consists of four specialized dashboard views organized in a structured 2x3 chart grid under a fixed widescreen resolution of  $1600 \times 900$ :

### 1. University Overview

- Focus: High-level institutional metrics and score distribution.
- KPIs: Total Universities (COUNTD), Global Average Score (AVG), Top Region Representation.
- Charts: Global Ranking Distribution, Score Bins Breakdown, Geographic University Heatmap, Regional Distribution, Top 10 Overall Universities, Category Breakdown.

### 2. Research Analytics

- Focus: Research volume, citation impact, and publication output.
- KPIs: Avg Research Score (AVG), Avg Citation Impact (AVG), Top Research Institution.
- Charts: Research Score vs. Academic Reputation, Citations vs. Global Score Scatter Plot, Top 15 Research Institutions, Country Research Strength, Avg Research Score Heatmap, Citations Distribution.

### 3. Student Analytics

- Focus: Diversity, international student enrollment, and faculty ratios.
- KPIs: Avg International Student % (AVG), Avg Faculty-to-Student Ratio (AVG), Top Diverse Campus.
- Charts: Faculty vs. International %, Top International Diversity, Faculty-to-Student Ratio Rankings, Global Student Diversity Map, Student Score vs. Global Rank, Top Diverse Campuses by Region.

### 4. Country Comparison

- Focus: Cross-national academic benchmarking and institutional volume analysis.
- KPIs: Total Countries Covered (COUNTD), Lead Nation in Research, Global Diversity Average (AVG).
- Charts: Country Research Strength, Country Diversity, University Volume by Country (COUNTD), Country Overall Metrics Matrix, Score Comparison by Nation, Regional Performance Summary.

### **Data Governance & Technical Standards**

- Aggregation Rules: All numeric metrics utilize non-additive aggregations (AVG for continuous scores/ratios and COUNTD for distinct institution and country counts) to eliminate data duplication across multi-level joins.
- Filter Governance: Data source filters permanently exclude Null values across key dimensions (Country, Score, University Name).
- UI/UX Consistency:
  - Header: Light Slate top banner across all views.
  - KPI Cards: White container background with 8px corner radius and light border padding.
  - Navigation: Standardized tab switching buttons with consistent positioning across sheets.

### **How to Access and Run**

1. GitHub Repository: Navigate to the Quainaat\_Rizvi branch at [https://github.com/springboardmentorpl/Higher-Education-Performance-Dashboard-Group-1/tree/Quainaat\\_Rizvi](https://github.com/springboardmentorpl/Higher-Education-Performance-Dashboard-Group-1/tree/Quainaat_Rizvi).
2. Tableau Workbook: Download eduvision\_dashboard\_v1.twbx located inside Module\_8\_Final\_Submission/ and open in Tableau Desktop 2023.1 or newer.
3. Presentation Mode: Press F7 to enter Presentation View. Ensure display resolution is set to 1600 × 900 for optimal grid alignment.
4. Interactivity: Use top navigation buttons to switch between dashboard tabs or click chart elements to trigger cross-dashboard filtering.